Broadband Omni-Directional Geophone

Built for High Resolution, High Efficiency, and High Reliability Seismic Acquisition with Less Operational Cost and Environmental Impact

Product Highlights

- Customized for China Oilfield Services Limited (COSL), Geophysical Acquisition and Surveying Division;
- Broadband (2Hz~190Hz) active force feedback Omni-Directional geophone with negative impedance compensation;
- Unique design provides ideal vector fidelity response for multi-component, high resolution ocean bottom/downhole seismic data recording and monitoring;
- Slight degree of background circuit noise (2uVrms@2~190Hz) for active force feedback sensor mode, providing ≥120dB instant dynamic range, which matches with most seismic acquisition systems;
- Low distortion (<0.1 % horizontal), low power consumption (<10mW), high sensitivity level(80V/m/s) ;
- Work in either active force feedback mode (Bandwidth 2~190Hz) or normal geophone mode (Bandwidth 15~190Hz), which is controlled by SW pin.

Pin Definition

1	SW	Mode control input, +2.5V for active mode, GND for normal mode
2	OUT-	Differential signal output
3	OUT+	Differential signal output
4	GND	Power & signal ground
5	-2.5V	Negative power supply
6	+2.5V	Positive power supply

Specifications

Active Mode

Equivalent Natural Frequency	2Hz±10%
Sensitivity	80V/m/s ± 10%
Spurious Frequency	> 190Hz
Output Resistance	2000 Ω
Distortion	<0.1 % horizontal, <0.4% vertical and pins-down
Maximum tilt angle	0°~360°
Equivalent Output noise	2uVrms@2~200Hz
Dynamic Range	120dB
Power Supply	±2.5VDC
Power Consumption	≤10mW
Diameter	26.7mm
Height	42mm
Weight	110g
Operating temperature	-40 to 100 °C

Normal Mode

Natural Frequency	15Hz ± 4%
Spurious Frequency	> 190Hz
Open-circuit Damping	0.35 (-4% ~ +7%)
Damping with 6.2kΩ shunt resistor	0.707 (-4% ~ +7%)
Sensitivity	70V/m/s (-8% ~ +4%)
Sensitivity with 6.2kΩ shunt resistor	56V/m/s (-8% ~ +4%)
Coil Resistance	1500 Ω ± 4%
Distortion	<0.1 % horizontal, <0.4% vertical and pins-down
Maximum tilt angle	0°~360°
Moving mass	9.7g
Maximum coil excursion	2mm (p.p.)
Operating temperature	-40 to 100 °C

* All parameters are specified at 22 °C (71.6 °F) in Upright position, unless otherwise specified.

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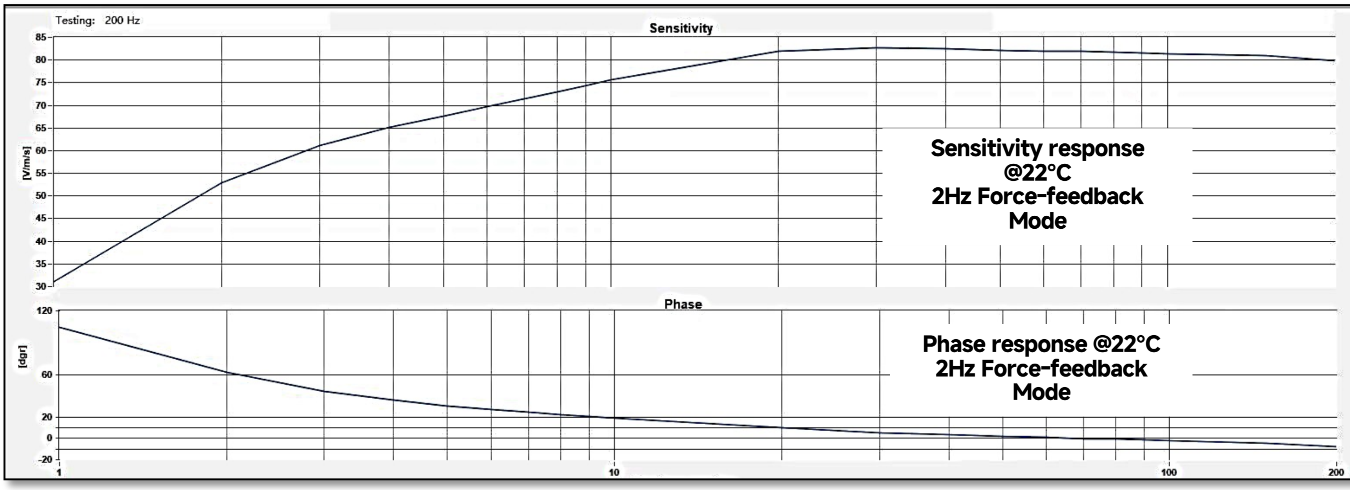
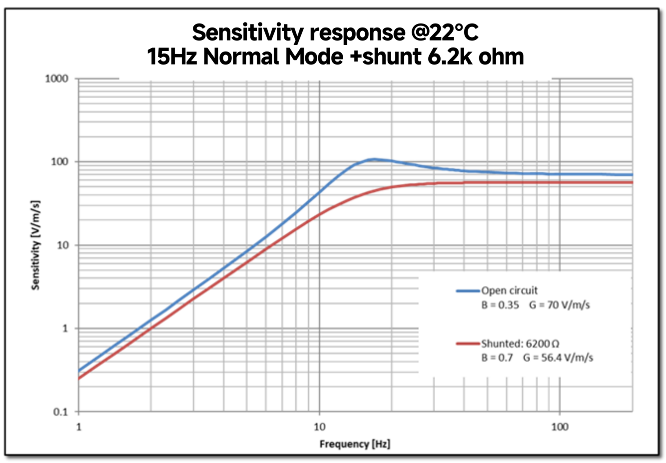
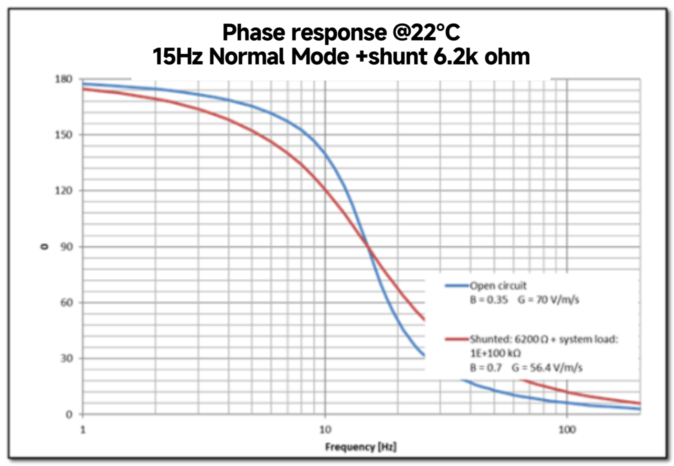
Sensitivity and Phase Response

Active Mode

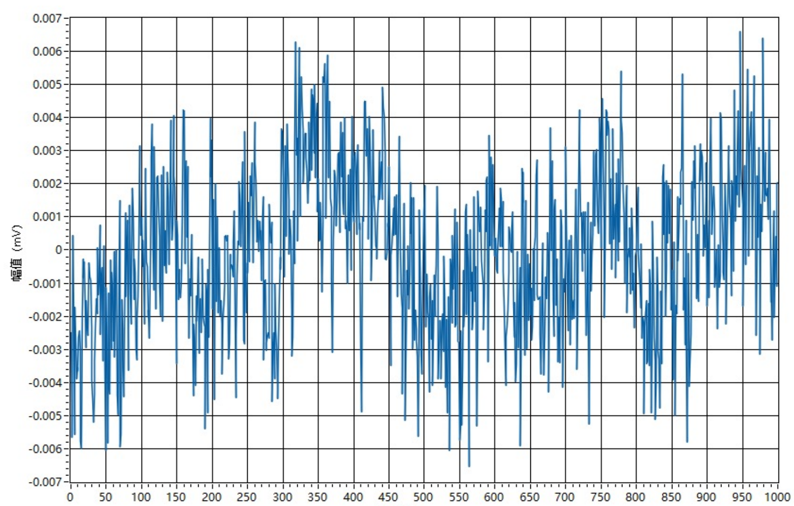


Normal Mode





Output Noise Level in Active Mode



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